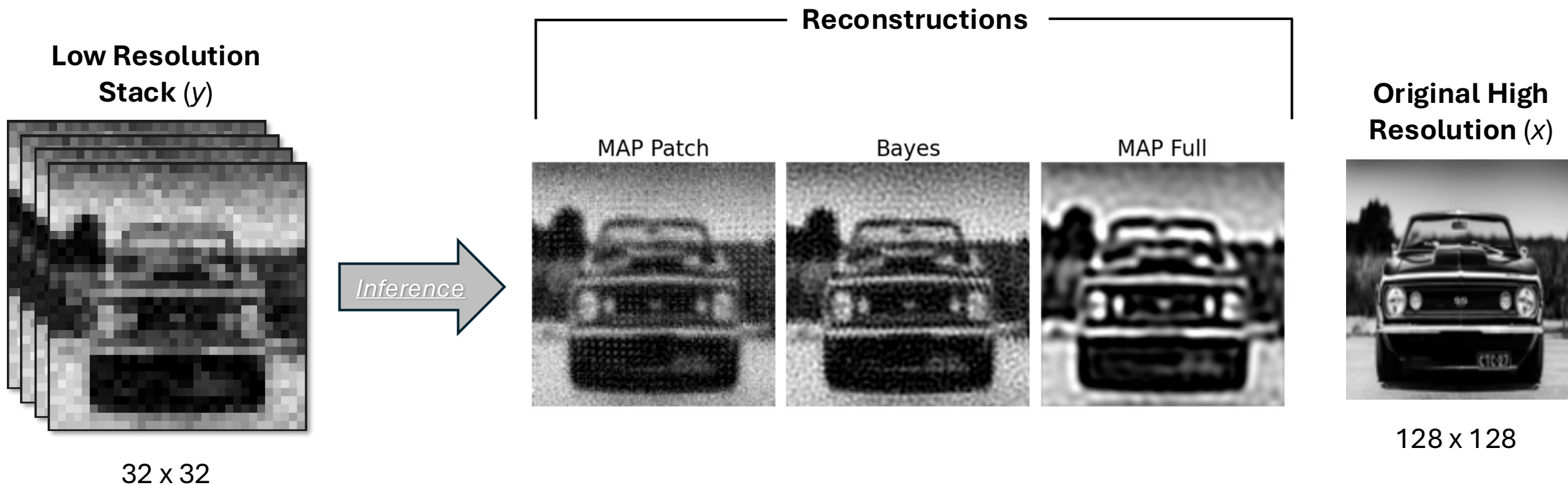


Bayesian Image Super-Resolution

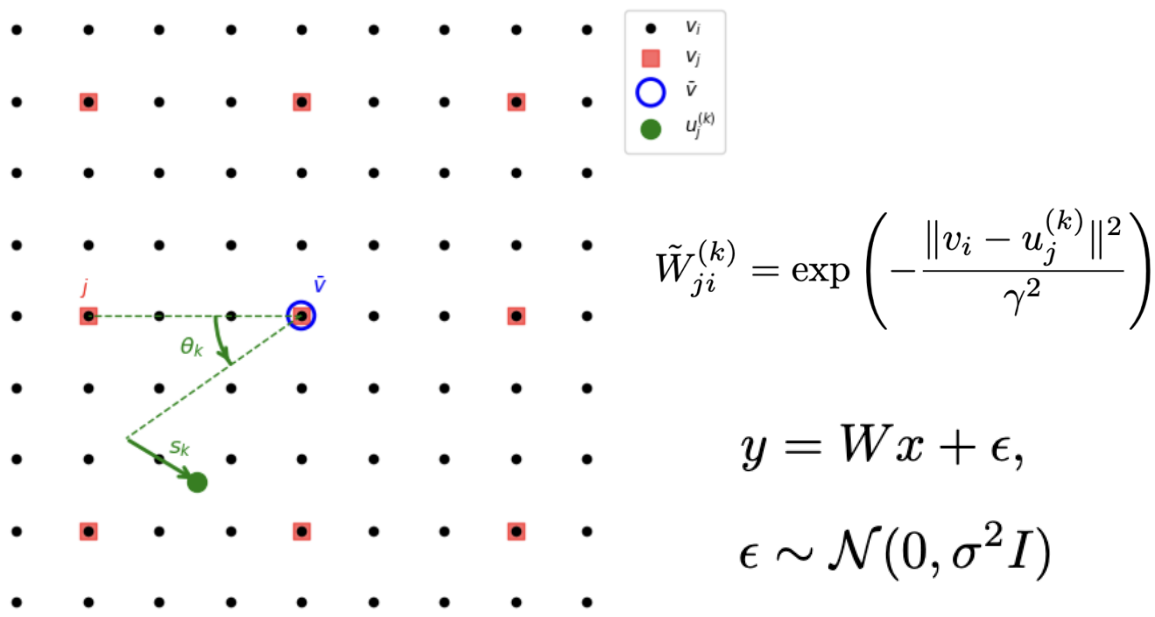


GOAL: Replicate the results of the paper & share the modular, open-source [implementation on GitHub](#).

Mathematical Framework

Create Low-Resolution Images (LR)

1. Downsampling
2. Rotation around center
3. Translation
4. Apply Point-Spread Function



Reconstruction Techniques

A] MAP

$$p(x, s, \theta, \gamma \mid y) \propto p(y \mid x, s, \theta, \gamma) \cdot p(x)$$

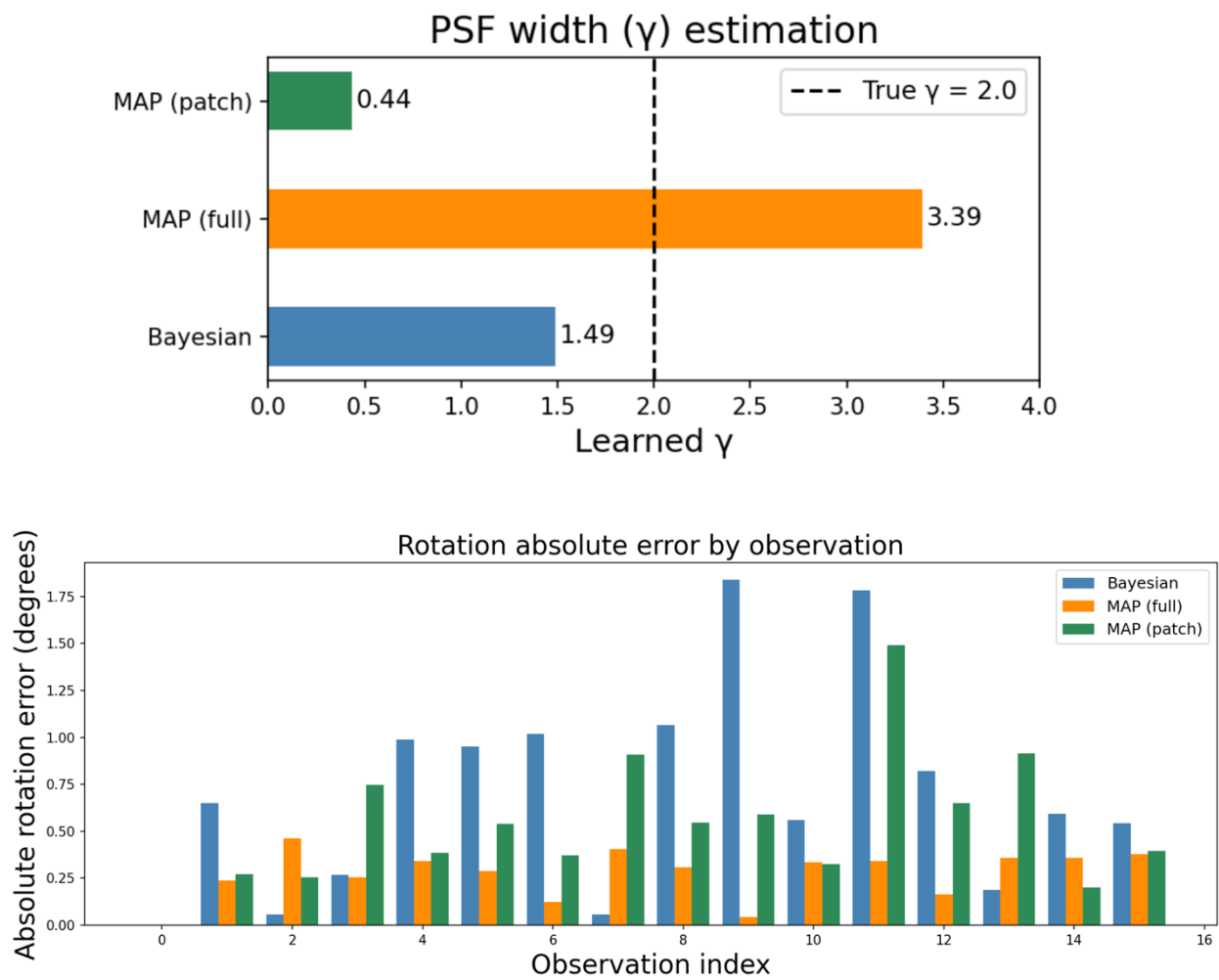
Gaussian process prior
↑

B] Bayes

$$p(y \mid s, \theta, \gamma) = \int p(y \mid x, s, \theta, \gamma) p(x) \, dx$$

- **Optimization:** PyTorch using L-BFGS optimizer
- **Patched Training:** To avoid memory overflow

Results



RESULTS: Ambiguous if Bayesian method truly outperforms MAP method, contrary to what the paper claims.

